

Full-Stack Web Applications with SAFE Stack

Lambda Days

About Me

- Software Engineer since 2011
- YouTube Content Creator
- .NET, Java, JavaScript



Goals

- Not another SAFE Stack Tutorial
- Very High Level
- Focuses on the Architectural Ideas
- There is some coding involved

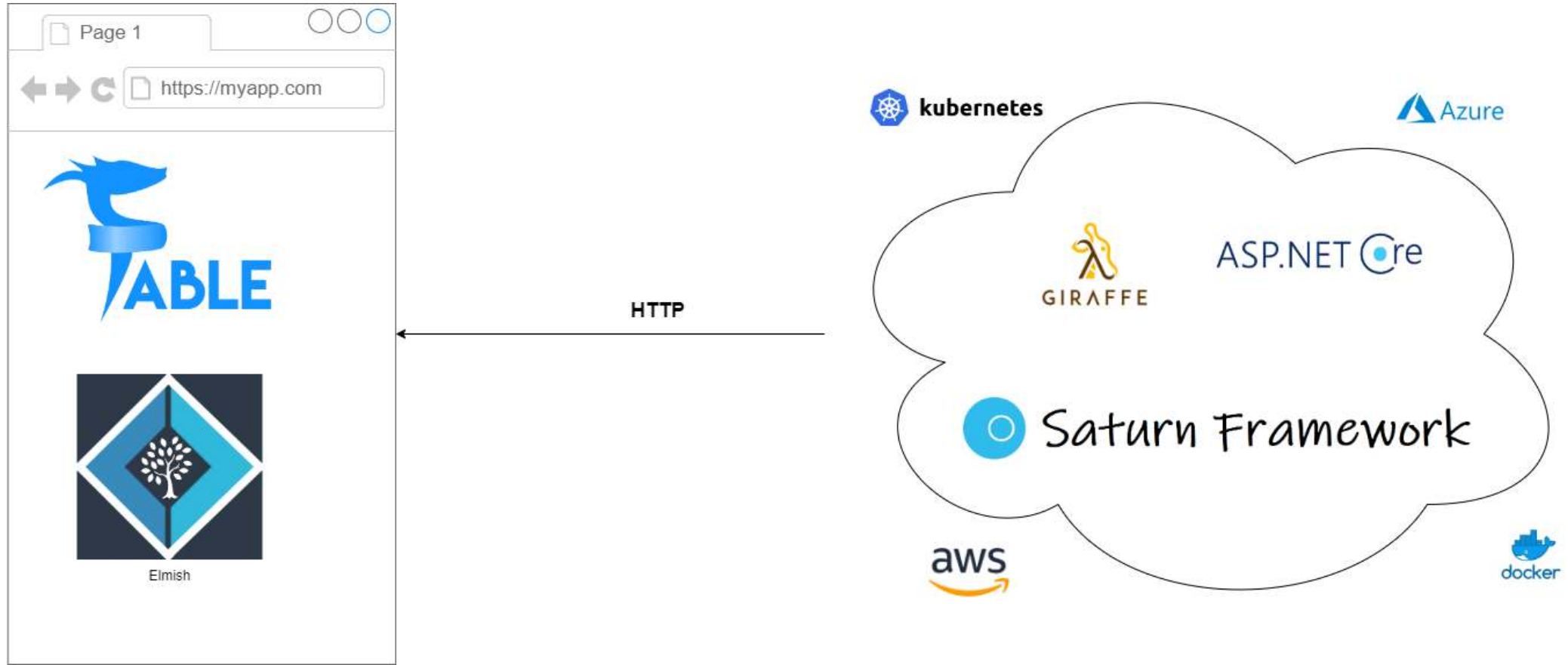
What is SAFE Stack?

- Full F# Web Stack (Client and Server)
- Functional First Architecture
- Strong Type Safety

What is the SAFE Stack?

- Suave/Saturn
- Azure/AWS
- Fable
- Elmish

What is the SAFE Stack?



Server

F# as a part of .NET Ecosystem

.NET Ecosystem

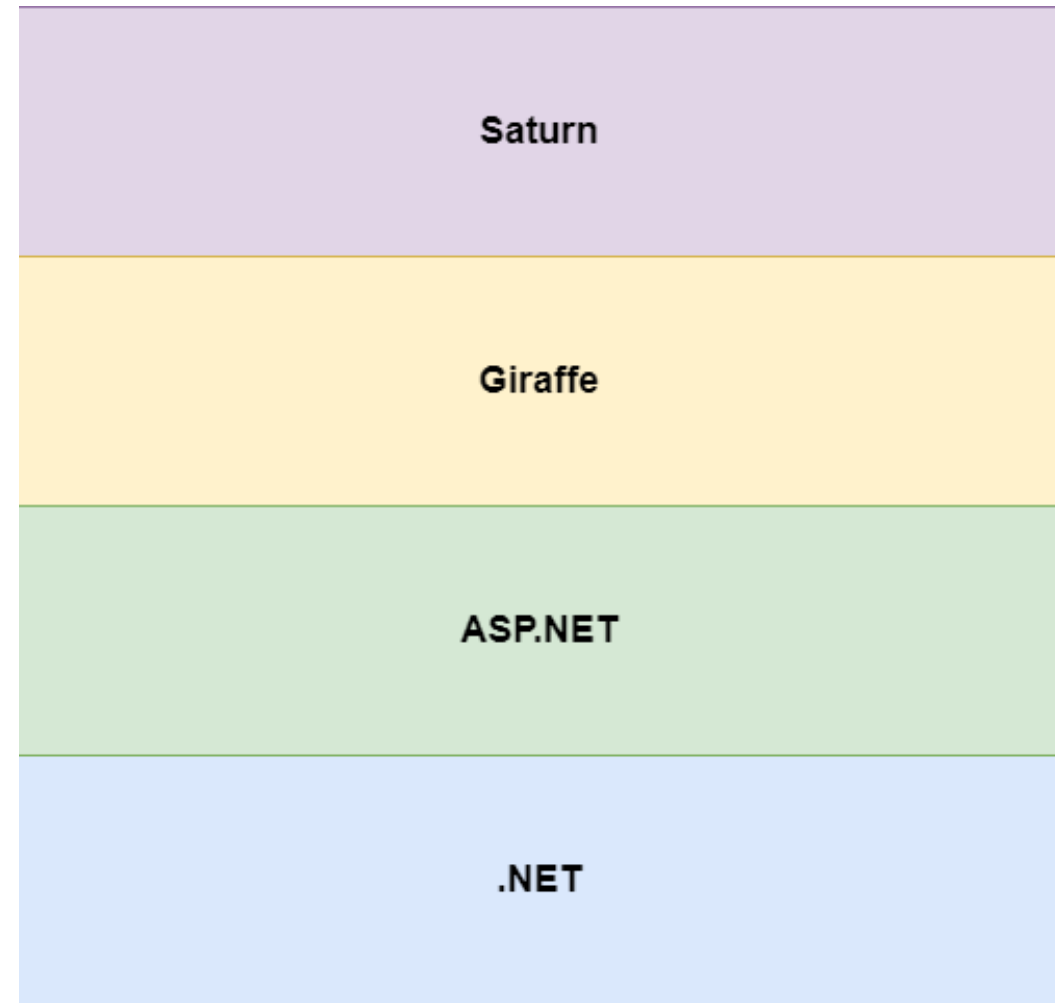
- Originally launched in 2002
- Was re-written in June 2016 as “.NET Core”
- Has been unified as just .NET
- Is fully cross platform and open source
- Is currently seeing a Renaissance

F# as Part of .NET

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- F# is an Integral part of .NET
- Ships with the .NET SDK out of the Box
- Reuses the same High-Performance Libraries and Tools

F# as Part of .NET

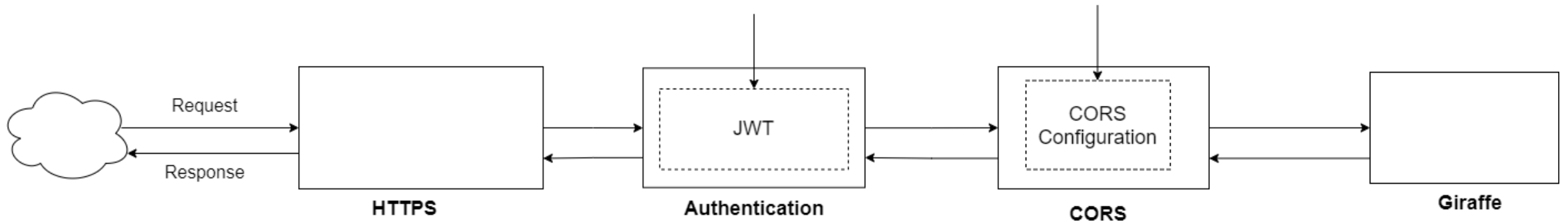


ASP.NET

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- High Performance .NET Web Framework
- Consists of a Pipeline Middleware and Services
- Runs a Production Ready Webserver called Kestrel

ASP.NET Middleware Pipeline



Giraffe

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- Simply an ASP.NET Core Middleware
 - Leverages a lot of the power of ASP.NET
 - Functional Architecture
- 
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Functional Architecture



HttpHandler



Web Application



```
type HttpFuncResult = Task<HttpContext option>
type HttpFunc = HttpContext → HttpFuncResult
type HttpHandler = HttpFunc → HttpContext → HttpFuncResult

let sayHelloWorld : HttpHandler =
  fun (next : HttpFunc) (ctx : HttpContext) →
    task {
      let greeting = sprintf "Hello World, from Nigeria"
      return! text greeting next ctx
    }
```



Giraffe
HttpHandler



```
let webApp =  
  choose [  
    route "/ping"    => text "pong"  
    route "/"        => htmlFile "/pages/index.html"  
    route "/hello"   => sayHelloWorld  
  ]
```



Handler Combination


```
let defaultView = router {
  get "/" (htmlView Index.layout)
  get "/index.html" (redirectTo false "/")
  get "/default.html" (redirectTo false "/")
}

let appRouter = router {
  not_found_handler (htmlView NotFound.layout) //Use the default 404 webpage
  forward "" defaultView //Use the default view
}

let app = application {
  error_handler (fun ex _ → pipeline { render_html (InternalError.layout ex)
}) use_router Router.appRouter
  url "http://0.0.0.0:8085/"
  memory_cache
  use_static "static"
  use_gzip
}

[<EntryPoint>]
let main _ =
  printfn "Working directory - %s" (System.IO.Directory.GetCurrentDirectory())
  run app
  0 // return an integer exit code
```



Saturn

Documentation

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- ASP.NET Core – <https://docs.microsoft.com/aspnet/core/>
 - Giraffe - <https://github.com/giraffe-fsharp/Giraffe>
 - Saturn - <https://saturnframework.org/>
- 
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Demo Time

Simple Giraffe Web Application

Browser

F# as a part of JavaScript Ecosystem

JavaScript Ecosystem

- Originally launched with the early web in 2002
- Experienced a Renaissance in 2008 due to V8 and Browser Wars
- In 2009, NodeJS was created making Server-Side Application possible
- Language has gone through several revisions
 - ES1 – ES6
 - ES2016 – ES2020
- Libraries and Frameworks Evolve Rapidly
- Modern JavaScript uses Build Tools like Webpack, Rollup etc.

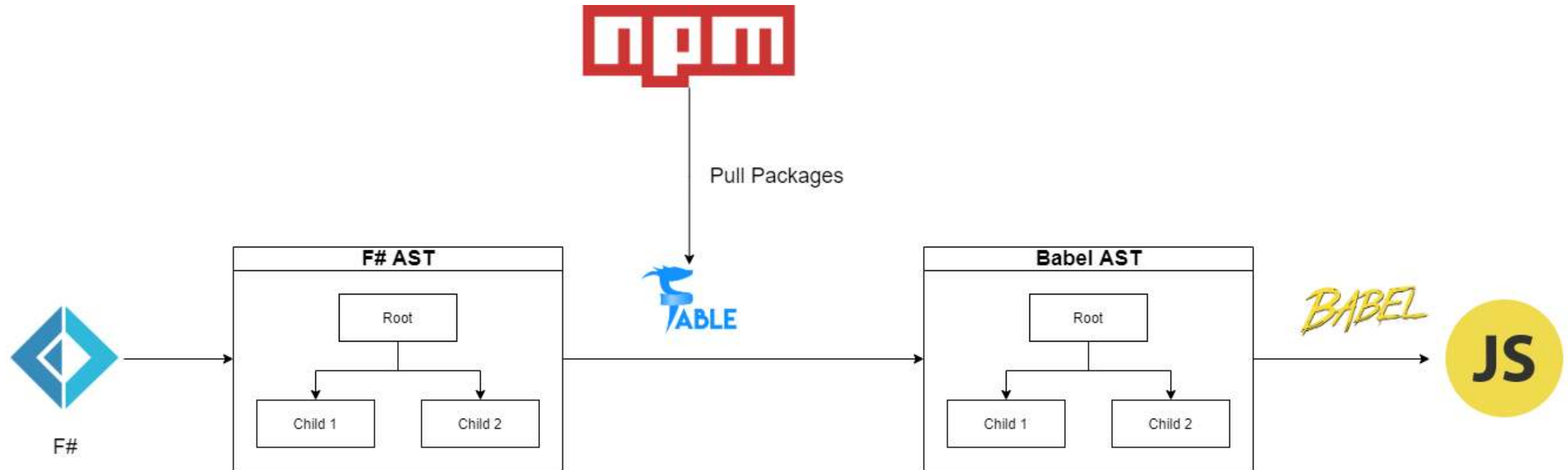
F# as Part of the JavaScript Ecosystem

- F# on the client is made possible by the Fable
- Fable is simply a Webpack plugin like LESS or SASS
- Reuses the same Client-Side Libraries and Tools like Webpack
 - Tree Shaking
 - Hot Module Replacement (Live Reload)
 - Minification
 - CSS Preprocessor etc.
- Web Assembly is also supported via Bolero Project
- Fable is also compatible with NodeJS

Fable

- Converts F# to JavaScript
- Uses the Babel JavaScript Compiler
- Shims out .NET APIs
 - Some API are replaced with JavaScript APIs e.g., Date
 - Some APIs are reimplemented in JavaScript e.g., Async
 - .NET Library Shims for Native JavaScript APIs
- Fully compatible with Existing NPM packages

FABLE Transpilation



FABLE Configuration

```
{
  "private": true,
  "scripts": {
    "start": "webpack-dev-server"
  },
  "dependencies": {
    "@babel/core": "^7.8.4",
    "fable-compiler": "^2.4.15",
    "fable-loader": "^2.1.8",
    "react": "^16.12.0",
    "react-dom": "^16.12.0",
    "webpack": "^4.41.6",
    "webpack-cli": "^3.3.11",
    "webpack-dev-server": "^3.10.3"
  }
}
```

webpack.config.js

```
var path = require("path");

module.exports = {
  mode: "development",
  entry: "./src/App.fsproj",
  output: {
    path: path.join(__dirname, "./public"),
    filename: "bundle.js",
  },
  devServer: {
    publicPath: "/",
    contentBase: "./public",
    port: 8080,
  },
  module: {
    rules: [{
      test: /\.fs(x|proj)?$/,
      use: "fable-loader"
    }]
  }
}
```

package.json

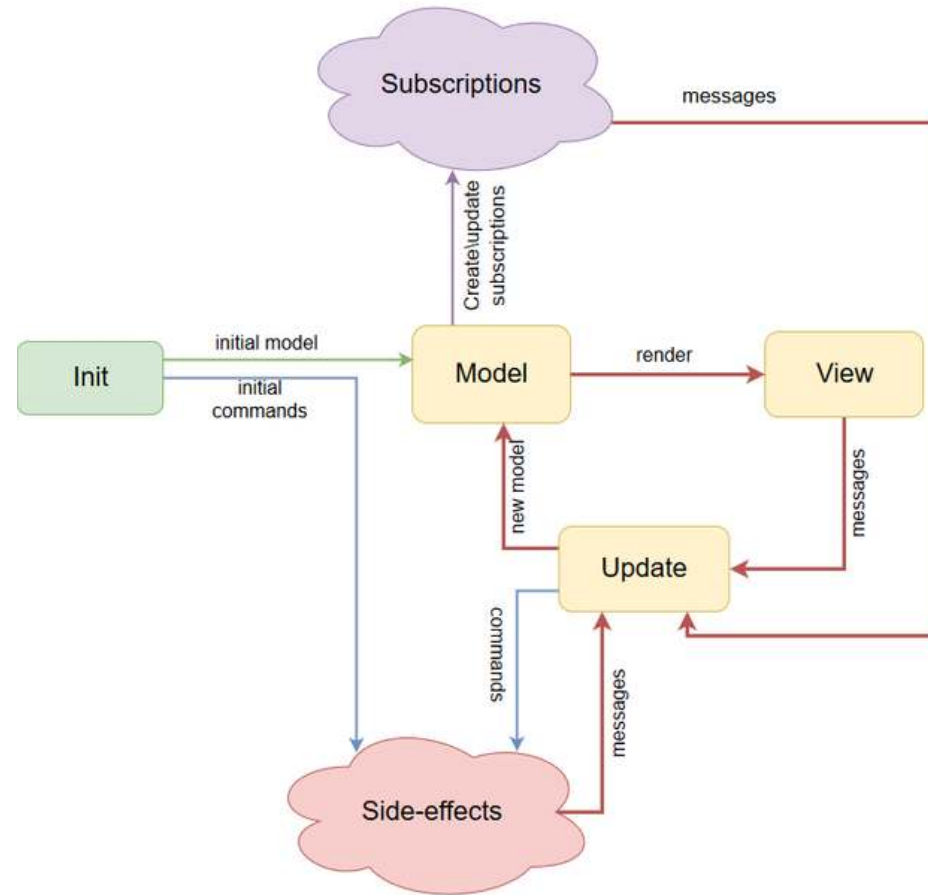
Demo Time

Simple Fable Application

Elmish

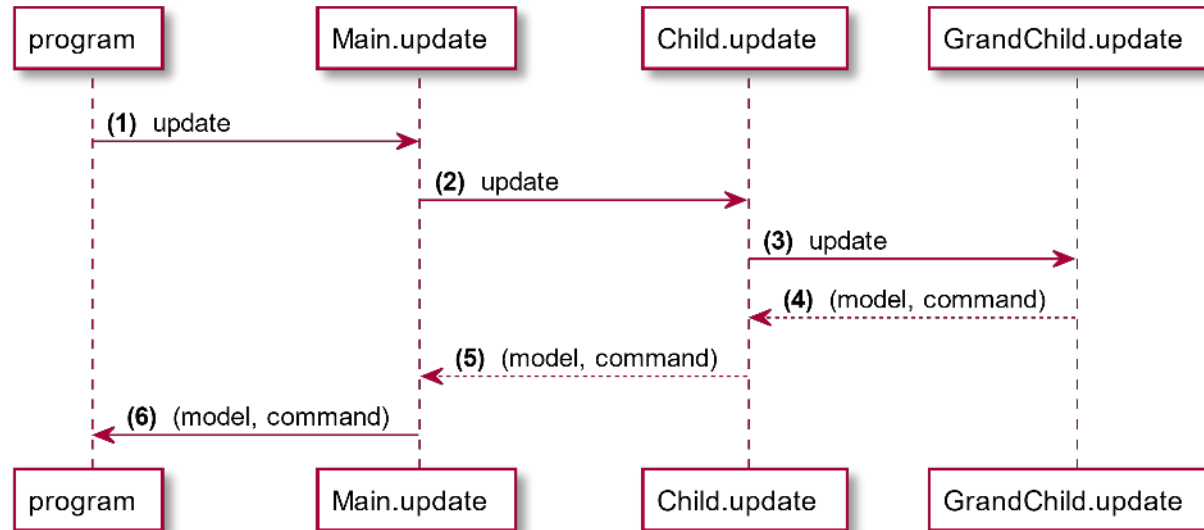
- Single Page Application Framework
- Leverages the ELM Architecture
- Uses React under the Hood
- Model-View-Update Pattern
 - Init – Creates the model
 - Update – Replaces the model in response to an Event (Message)
 - View – Renders the UI

Elm Architecture



Source: <https://steemit.com/utopian-io/@tensor/using-the-elm-architecture-or-the-mvu-pattern-with-dartea-inside-of-dart-s-flutter-framework>

Elmish Composition



Source: <https://elmish.github.io/elmish/>

Documentation

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- Fable – <https://fable.io/docs/>
- Elmish Book - <https://zaid-ajaj.github.io/the-elmish-book/>

SAFE Stack

Putting it all together with a nice bow on top

SAFE Stack

- Install using *dotnet new -i SAFE.Template*
- Documentation available at <https://safe-stack.github.io/>
- Enterprise Support Available by Compositional.IT

Demo Time

SAFE Stack Example – Edelwiess Data

Thank You

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youtube.com/odytrice